

Assignment 10

Name: _____

Student ID: _____

11 problems. Show all of your work.

Problem 1.

Find the absolute maximum and minimum values of

$$f(x, y) = x^2 - xy + y^2 - 3y$$

on the region bounded by the x - and y -axes and the line $x + y = 4$.**Problem 2.**Find and classify all the critical points of $f(x, y) = xye^{x^2+y^2}$.**Problem 3.**Find the maximum and minimum values of the function $f(x, y) = 2x^2 + 3y^2 - 4x - 5$ on the domain $x^2 + y^2 \leq 121$.**Problem 4.**

The business manager for Acme Corporation plots a grid on a map of the region Acme serves and determines that the company's three most important customers are located at points $A(1, 5)$, $B(0, 0)$, and $C(8, 0)$, where units are in miles. At what point $W(x, y)$ should a warehouse be located in order to minimize *the sum of the squares of the distances* (NOT the square of the sum) from W to A , B , and C ?

Problem 5.Find all local maxima, minima and saddle points for $f(x, y) = \frac{1}{3}x^3 - x + xy^2$.**Problem 6. Tangent Planes Parallel to a Given Plane**

Find the points on the surface

$$x^2 + y^2 + 3z^2 = 1$$

at which the tangent plane is parallel to the plane $-x - 5y - 2z = -3$.

Problem 7. Maximizing a Cobb-Douglas utility

A consumer has \$600 to spend on two commodities, the first of which costs \$20 per unit and the second \$30 per unit. Suppose that the utility derived by the consumer from x units of the first commodity and y units of the second commodity is given by the Cobb-Douglas utility function

$$U(x, y) = 10 x^{0.6} y^{0.4}.$$

How many units of each commodity should the consumer buy to maximize utility?

Problem 8.

You have 24 square inches of cardboard and want to build a box in the shape of a rectangular prism. Find the dimensions of the box, if you would like it to have the largest possible volume. See Figure 1.

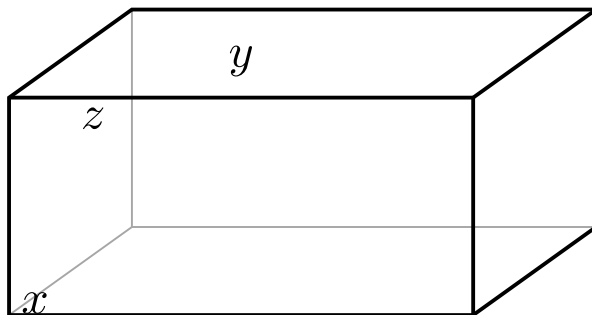


Figure 1. A rectangular prism with dimensions x , y , and z .

Problem 9.

Use the method of Lagrange Multipliers to find the maximum and minimum values of $f(x, y) = \sqrt{2}x - \sqrt{2}y$ subject to the constraint $x^2 + y^2 = 4$.

Problem 10.

Given $f(x, y) = -x^2 + 4x - y^2 - 4y$.

- A. Find the maximum and minimum values of $f(x, y)$ on the circle $x^2 + y^2 = 9$, first using Lagrange multipliers, then again by parametrizing the circle.
- B. Find the maximum and minimum values of $f(x, y)$ over the disk $x^2 + y^2 \leq 9$.

Problem 11.

You are in charge of erecting a radio telescope on a newly discovered planet. To minimize interference, you want to place it where the magnetic field of the planet is weakest. The planet is spherical, with a radius of 6 units. Based on a coordinate system whose origin is at the center of the planet, the strength of the magnetic field is given by

$$M(x, y, z) = 6x - y^2 + xz + 60.$$

Where should you locate the radio telescope?